Worksheet 4-2 A Straight-Line Motion #1
AP Calculus Name_____

Directions: The following problems involve straight line motion. Unless otherwise stated, the functions modeling the position, velocity, and acceleration are continuous and differentiable.

1. For $t \geq 0$, a particle moves along the x axis with velocity $v(t) = t^2 - 8t + 12$. It is known that $x(0) = -5$

(a) Find the acceleration of the particle when $t = 10$.

(b) Find all times when the particle is at rest.

(c) For what times is the particle moving to the right?

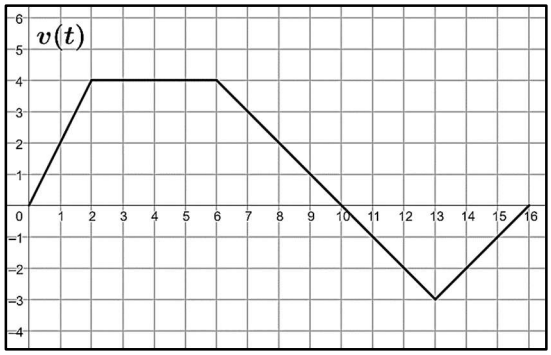
(d) Find $a(3)$. Is the particle speeding up or slowing down at $t = 3$? Give a reason for your answer.

(e) Is the particle moving towards or away from the origin at $t = 0$? Give a reason for your answer.

(f) What is the velocity of the particle when its acceleration is 0?

(g) What is the acceleration of the particle when $t = 0$?

(h) At what time(s) is the velocity of the particle equal to 60?



2. For $0 \leq t \leq 16$ seconds, a bug is walking along a straight path. The velocity of the bug can be modeled by $v(t)$ as shown in the figure above, where $v(t)$ is measured in centimeters per second.

(a) At what time t did the bug turn around? Give a reason for your answer.

(b) Find $a(14)$. Include units of measure.

(c) On what intervals does the bug have a positive acceleration? Give a reason for your answer.

(d) Is the bug speeding up or slowing down when $t = 7$? Give a reason for your answer.

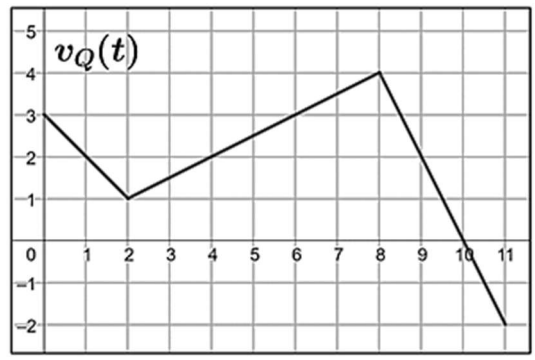
(e) Is the bug speeding up or slowing down when $t = 11$? Give a reason for your answer.

(f) At what time(s) is the speed of the bug equal to 2?

(g) What is the velocity of the bug when $a(t) = 0$?

(h) A second bug is also walking along the path. The position of the second bug is given by $B_2(t) = 6\sin\left(\frac{\pi t}{3}\right) - 2t$. Which bug has a greater velocity when $t = 1$?

t	1	4	6	7	11
$x_P(t)$	-3	12	9	0	1
$v_P(t)$	2	5	-8	-1	3



3. For $0 \le t \le 11$, particles P and Q are moving along the x axis. The positions of the particles are x_P and x_Q respectively. Selected values of the position and velocity for particle P are given in the table above along with a graph of the velocity for particle Q. It is known that $x_Q(6) = 5$, and that $x_p(t)$ and $v_p(t)$ are both continuous.

(a) Approximate $a_P(5)$. Show the calculations that lead to your answer.

(b) Find $a_Q(\pi)$.

(c) At what time(s) t does particle Q turn around? Give a reason for your answer.

(d) Are particles P and Q moving toward or away from each other at time $t = 6$? Give a reason for your answer.

(e) Is particle Q speeding up or slowing down when $t = 1$? Give a reason for your answer.

(f) Can you guarantee a time c, where $1 < c < 11$, such that $v_P(c) = 0$? Justify your answer.

(g) What is the velocity of particle Q when $x_P(t) = 9$?

t (seconds)	0	$0 < t < 4$	4	$4 < t < 9$	9	$9 < t \leq 15$
$x(t)$ (meters)	15	???	-4	negative	0	positive
$v(t)$	???	negative	0	positive	0	positive
$a(t)$	2	???	3	???	-4	negative

4. For $0 \leq t \leq 15$, a particle moves along a horizontal axis with position $x(t)$, where $x(t)$ is measured in meters and t is measured in seconds. Information about the particle's position, velocity and acceleration is given in the table above with differentiable functions $x(t)$, $v(t)$ and $a(t)$.

(a) Use the data in the table to estimate $v(2)$. Include units of measure.

(b) Do the data in the table support the claim that there is a time c, for $0 < c < 15$, such that $a(c) = 0$? Justify your answer.

(c) Is the particle speeding up or slowing down when $t = 11$? Give a reason for your answer.

(d) At what time(s) does the particle change direction? Give a reason for your answer.

(e) Is the particle moving toward or away from the origin when $t = 6$? Explain your reasoning.

(f) In which direction is the particle moving when $t = 8$? Give a reason for your answer.

(g) For $0 \leq t \leq 4$, the acceleration of the particle can be modeled by $a(t) = 2^t + K - \sqrt{t} - Lt$, where K and L are constants. Find K and L.

5. (**Calculator Active**) For $0 \leq t \leq 3$, the velocity of a particle moving along the x axis is given by the equation $v(t) = 1.3t^2 - 2.5t^{1.3} + \sin(t^2)$. It is known that $x = -4$ when $t = 2$.

(a) Find $a(1)$.

(b) Is the particle speeding up or slowing down when $t = 2$? Give a reason for your answer.

(c) Is the particle moving toward or away from the origin when $t = 2$? Give a reason for your answer.

(d) At what time(s) in the interval $(0, 3)$ does the particle change directions? Explain your reasoning.

(e) Find all times in the interval $(0, 3)$ when the particle is moving to the right? Give a reason for your answer.

(f) For $0 \leq t \leq 3$, at what time(s) is $v(t) = -1$?

(g) Find all times in the interval $(0, 2)$ when $a(t) = 0$.

(h) For what times in the interval $(0, 3)$ does the particle have positive acceleration?

(i) What is the acceleration of the particle when $v(t) = 1$?

(j) Find all time(s) in the interval $(0, 2)$ when the speed of the particle is 1.5.

6. A rock thrown vertically upward from the surface of the moon at a velocity of 32 meters per second reaches a height of $s(t) = 32t - 0.8t^2$ meters in t seconds.

a. Find the rock's velocity and acceleration as functions of time.

b. How long did it take the rock to reach its highest point?

7. The data in the table gives selected values for the velocity, in meters per minute, of a particle moving along the x-axis. The velocity v is a differentiable function of time t.

Time t	0	2	5	6	8	12
Velocity $v(t)$	–3	2	3	5	7	5

a. At $t = 0$, is the particle moving to the right or left? Justify.

b. Is there a time during the time interval $0 \le t \le 12$ minutes when the particle is at rest? Justify.

c. Use the data from the table to approximate $v'(10)$.

d. Explain the meaning of $v'(10)$ in terms of the particle's motion.

8. A particle moves along the x-axis so that at time t its position is given by $x(t) = t^3 - 6t^2 + 9t + 11$ where t is measured in seconds and x is measured in feet.

a. At $t = 0$, is the particle moving to the right or left? Justify

b. At $t = 1$, is the velocity of the particle increasing or decreasing? Justify.

c. What is the displacement over the first 6 seconds?